

Unit 9. Futures Directions in SM

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Unit 9. Future Directions in SM

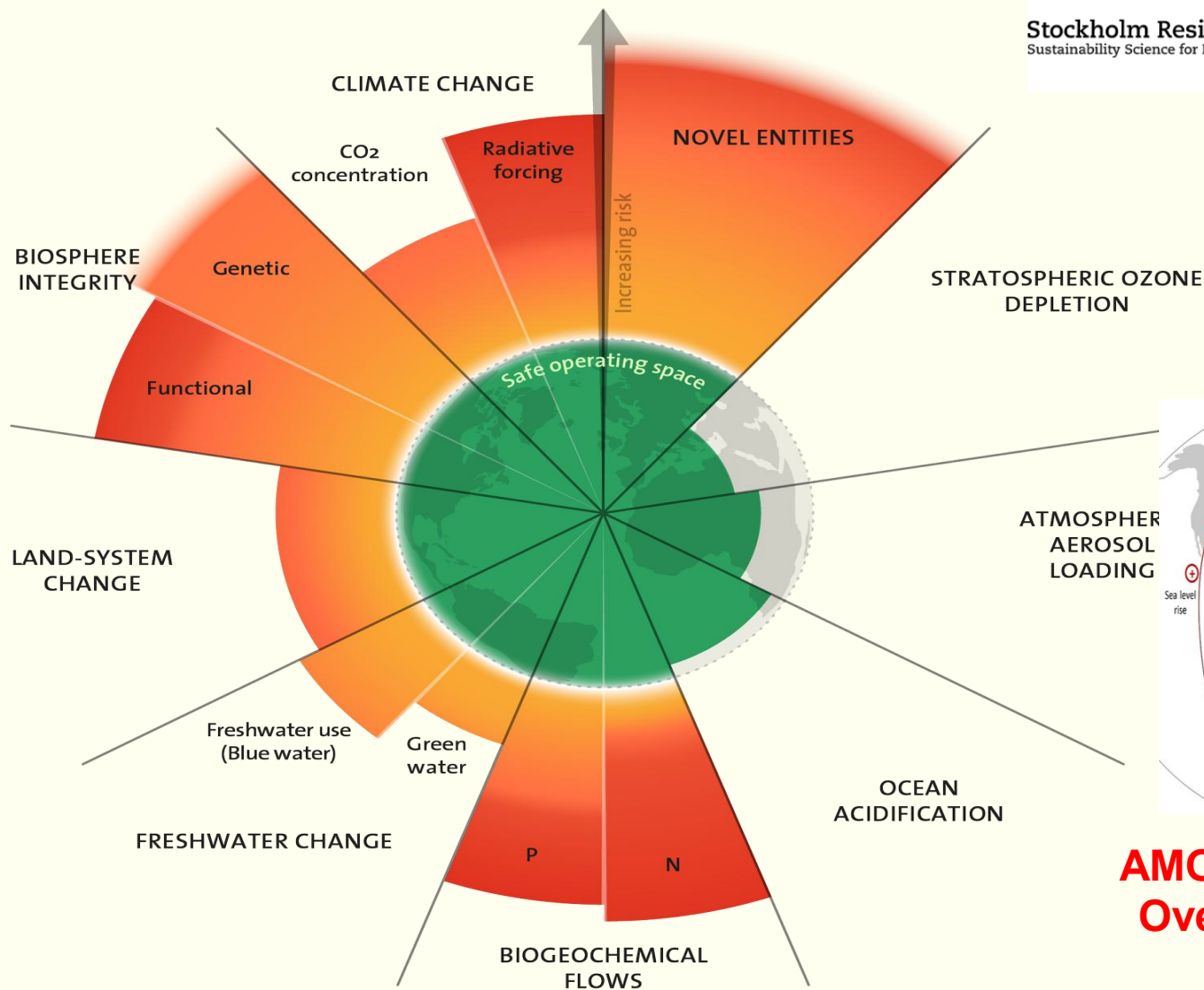
LEARNING OBJECTIVES

1. To understand the current climate emergency
2. To discover the main challenges of Environmental Sustainability for Strategic Management
3. To understand the current logic of B-A-S
4. To discover the Regenerative Strategy

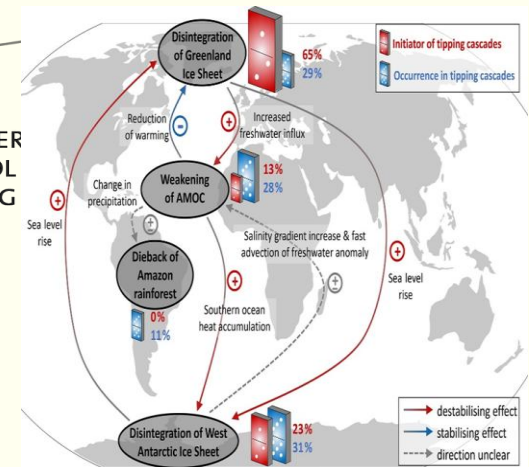


9.1. The Current Climate Emergency

(1) The “Nine Planetary Boundaries”



Stockholm Resilience Centre
Sustainability Science for Biosphere Stewardship



AMOC: Atlantic Meridional Overturning Circulation

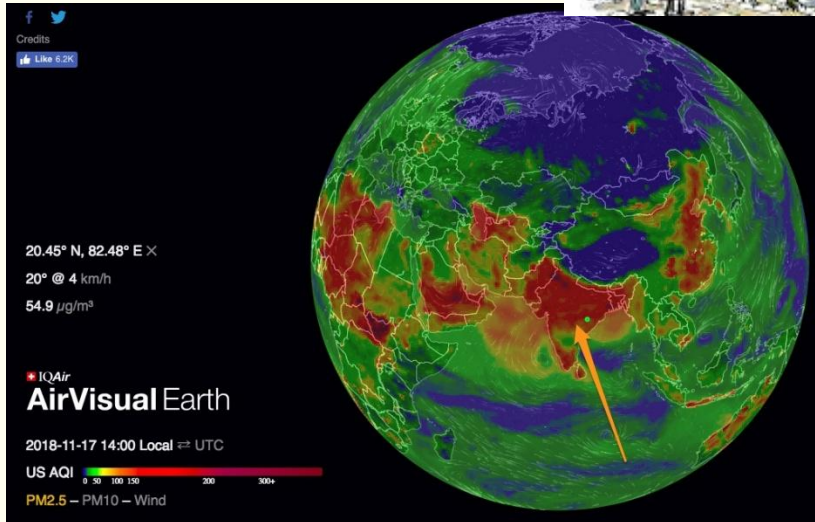
9.1. The Current Climate Emergency

1. Extreme cases of the "Tragedy of the Commons" (Hardin, 1968)

Ocean Pollution...



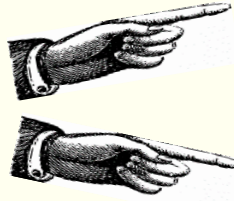
....& Atmosphere Pollution



9.2. Main Challenges for SM

1. Climate Emergency: The Grand Challenge

Tragedy of the Commons



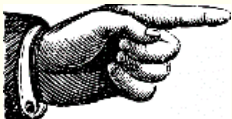
Disruptive technology....

Change in Moral Sentiments....

1. To reconsider Business & Sustainable Development systemic and ecological foundations (*Whiteman et al., 2013*)
2. Increasing expectations on businesses (*Alvarez et al., 2020*)
3. Overcome the 'Divorce' between Climate Science and Management Studies (*Nyberg & Wright, 2021*)

In 2019: + 11,000 scientific papers on CCU

4. There is no much time! (one/two decades) (*IPCC, 2023*)
5. Emission Reduction & Atmospheric CO₂ capture (*IPCC; 2023*)



9.2. Main Challenges in SM



But,...

What have we actually done?

9.2. Main Challenges in SM

Unit 2. Objectives and Values

Company's Mission & vision → Include the Nature!

A New Managerial Conception:

From Utilitarianism → Eco—Ecomotional Wealth

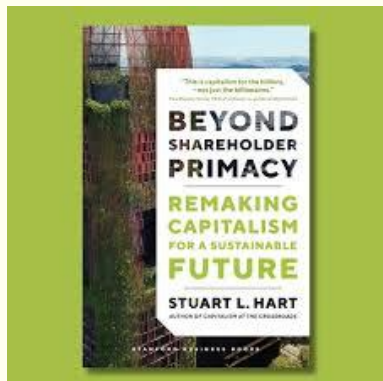
Performance = Success?

Going beyond ECONOMIC PROFIT & SHAREHOLDER VALUE CREATION

→ Environmental Performance, Reduce Footprint, REDUCE

*ENVIRONMENTAL NEGATIVE EXTERNALITIES → EVEN: RON (Return
On Nature?)*

TO PASS FROM SHAREHOLDERS' PRIMACY → STAKEHOLDERS MANAG.

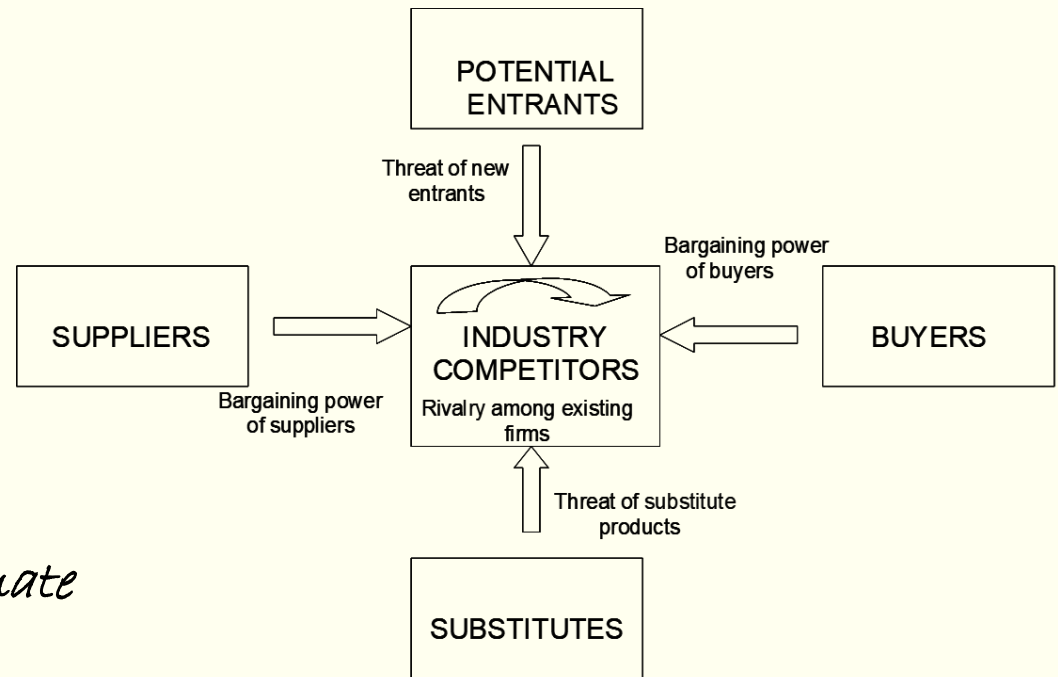


9.2. Main Challenges in SM

Unit 3. Business Environment

PORTER'S FIVE FORCES MODEL

PESTEL ANALYSIS

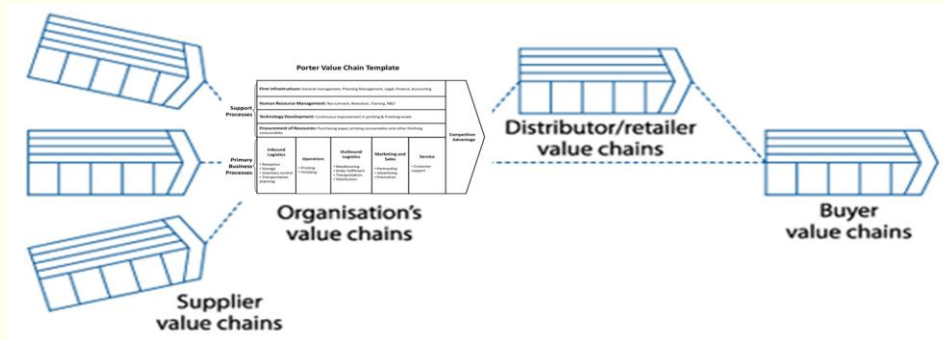


*What Force is more
Challenging/Affected by Climate
Emergency?*

Source: Porter (1980).

9.2. Main Challenges in SM

Unit 4. Internal Analysis



What's New?

STRENGTHS

WEAKNESSES

OPPORTUNITIES

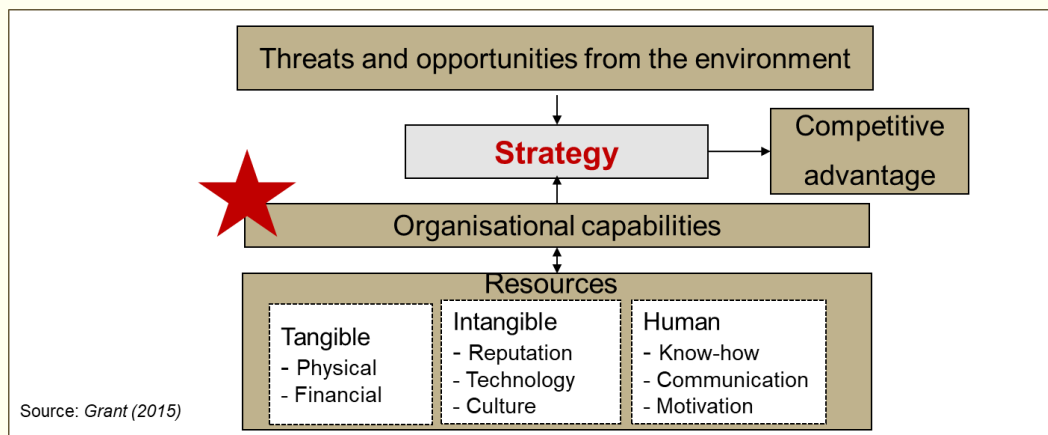
THREATS

MAINTAINING

CORRECTING

EXPLOITING

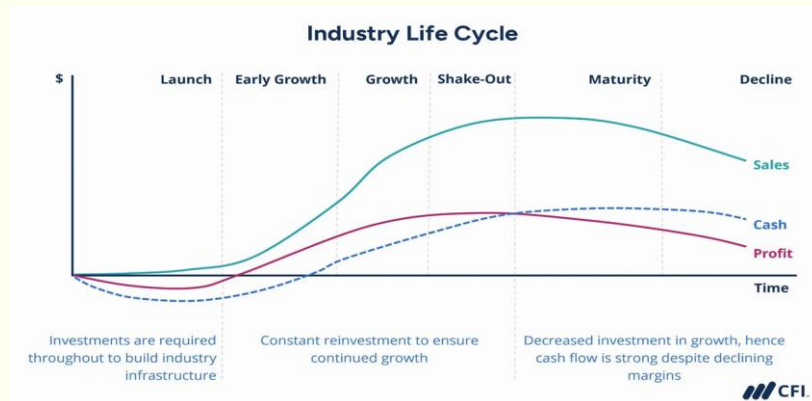
AVOIDING



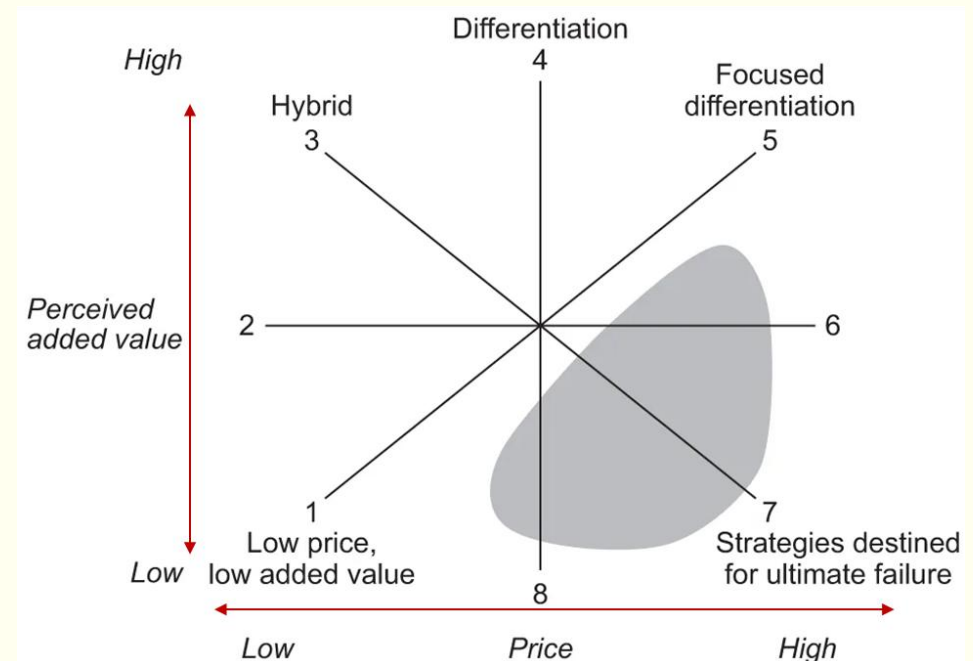
9.2. Main Challenges in SM

Unit 5. Competitive Strategy

What means "Sustainable Business Models?"



The "STRATEGY CLOCK"



9.3. Business-As-Usual



ENVIRONMENTAL STRATEGIES

1. *'Just Environmental Integration' ... "Greening the business"*
2. *'Normative & Stakeholders' Pressures' (Delmas & Toffel, 2004)*
3. *'Green & Competitive?' (Porter & van der Linde, 1995) 'Short-Term Orientation & Shareholders Primacy' (Orsato, 2006)*
4. Aim: Ignorance-Reduction-Minimization of Pollution (Hart, 1995)

9.3. Business-As-Usual

2. Business-As-Usual: Denial

- *The Brief history of (**Un**)Sustainable Research'* (Ergene 2021)
- 'Different ways of '**denial**' (Nyberg & Wright, 2021)
 1. **Literal** → Climate change/Anthropocentrism rejection
 2. **Implicatory** → Climate change, NOT climate emergency
 3. **Interpretative** → Current business models/tools can manage climate crisis

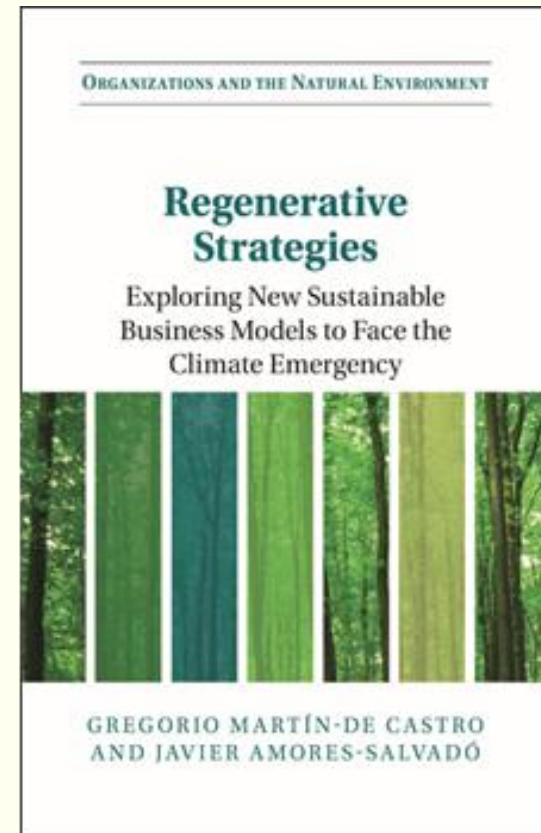


Disruptive technology....



Change in Ethic & Moral Sentiments....

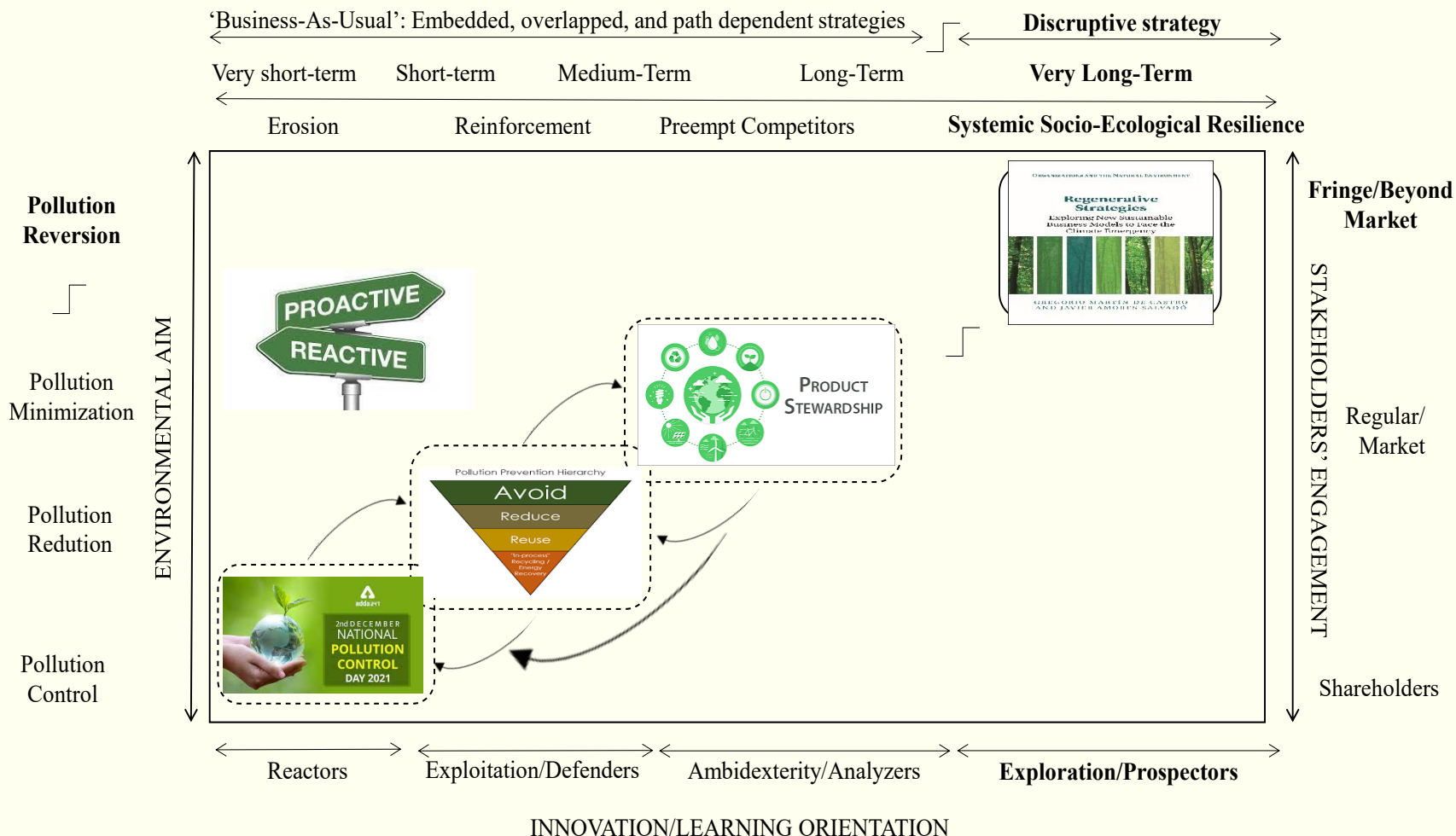
9.4. The Regenerative Strategy



9.4. The Regenerative Strategy

Regenerative Strategy vs. B-a-U Logic

COMPETITIVE POSITIONING/ORIENTATION (LOGIC, TIMING, AND PURPOSE)



9.4. The Regenerative Strategy

Regenerative Strategy

Goes beyond zero-pollution logic and seeks, as an environmental aim, the reversion of current environmental degradation and the creation of positive environmental externalities, which implies fundamental changes in the logic of doing business.

Disruptive technology

Atmospheric carbon dioxide capture and utilization (sequestration, feedstock for other products, ...)

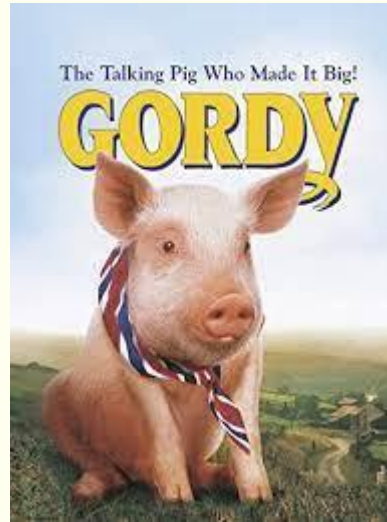
Change in Ethical and Moral Sentiments

EcoEmotion Wealth → Environ. Perf → Systemic Socio-Eco. Resilience

INDIVIDUAL → ORGANIZATION → THE PLANET



9.4. The Regenerative Strategy



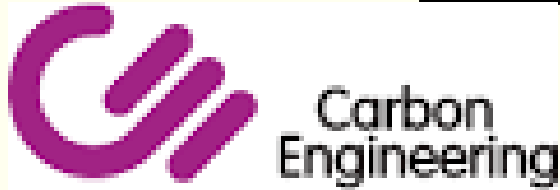
The case:

Carbon Engineering

9.4. The Regenerative Strategy

4. The Case of Carbon Engineering

(1) Disruptive Technology → Climate Emergency



*“We believe humanity can
solve Climate Change”*

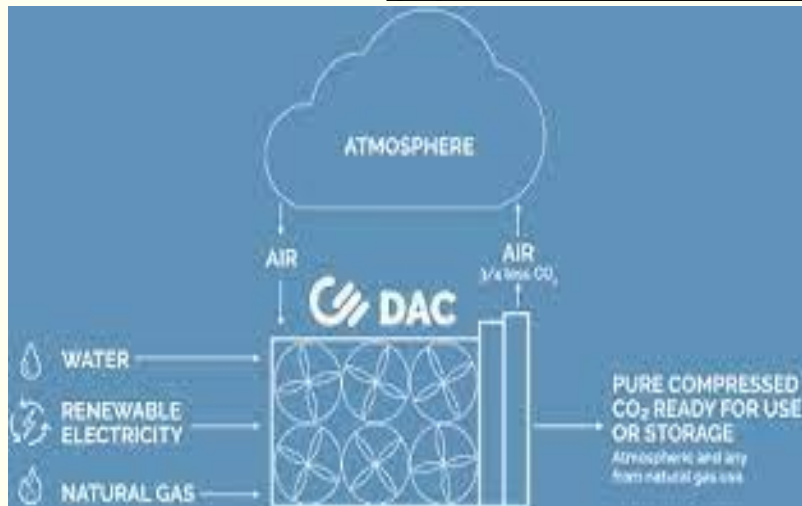
- Founded in 2009 in Calgary by Harvard Professor David Keith
- In 2013, the first DAC prototype plant (FOAK First of A Kind)
- **Mission** → *“To develop and commercialize a technology that captures CO₂ directly out of the atmosphere at **megaton-scale**”*
- **Business Model** → *“To license our technologies to local plant developers and energy industry partners around the world. It gives us the ability to deploy solutions as quickly and broadly as possible and to start making a meaningful impact on the climate challenge”*

9.4. The Regenerative Strategy



4. The Case of Carbon Engineering

(1) Disruptive Technology → Climate Emergency



DAC Direct Air Capture

1. **Megaton-scale** → DAC plant would capture 1 million tons of CO₂ per year
2. **Scalability** → Compatible with known supply chains and reliable equip. cost
3. **Affordable** → Approx US\$100 per ton at a large scale
4. **Freedom of location** → Due to the uniform distribution of CO₂
5. **Closed chemical cycle**
6. **Free emissions** → Achieving **net zero emissions, & Net-Negative ones!**

9.4. The Regenerative Strategy

4. The Case of Carbon Engineering

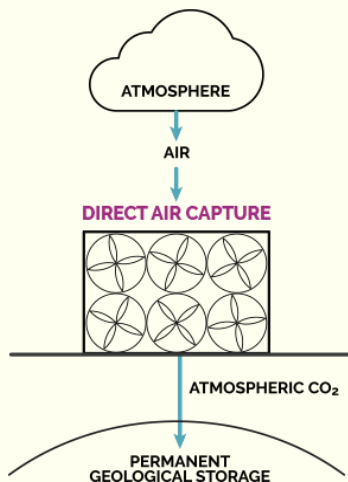


Carbon
Engineering

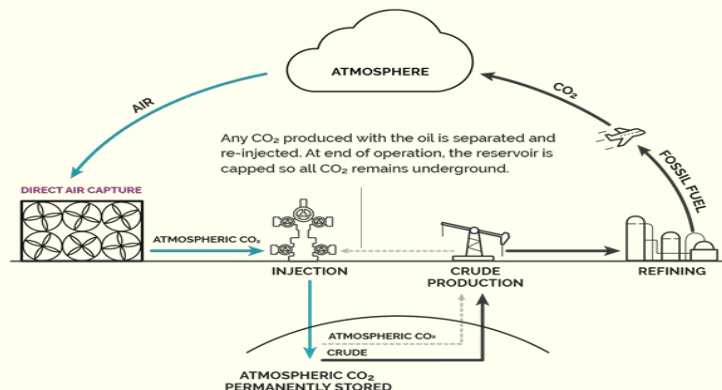


DAC Uses

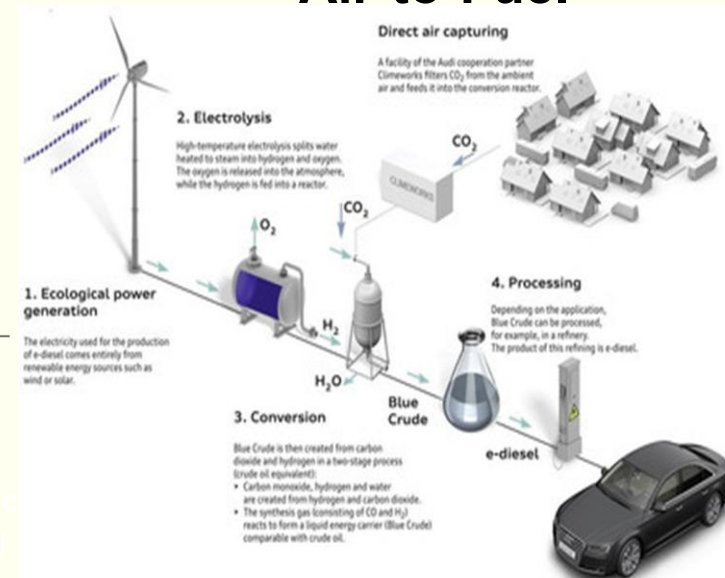
Soil Storage



Enhanced-Oil-Recovery



Air-to-Fuel



9.4. The Regenerative Strategy

4. The Case of Carbon Engineering

(2) New Purpose → Climate Emergency

*In-depth interviews, corporate web
archival data, scientific publications
on CCU and CE, dB*



David Keith, CE Founder and
Harvard Professor

**Eco-
Emotional
Wealth**

- ✓CE Mission, Vision → We believe humanity can solve climate change
- ✓Keith (2013:XVI-XVIII) → ***“A fuzzy love of nature is uncontroversial (...) For me, the challenge is to craft an environmental ethic that recognizes non-utilitarian values in the natural world without asserting that these values trump all others without making naïve claims of a sharp distinctions between nature and civilization”***
- ✓“Professor Keith is researching tools that are inmodest in the extreme: Planet-size ideas for Planet-size problems” (The Times magazine, Morton, 2009)

9.5. Towards a New Capitalism?

Anthropocentrism

